

Day 2 Assignment — Titanic Dataset Analysis

Use the Titanic dataset and perform the following tasks.

Part A — NumPy Tasks

1. Create a NumPy array containing marks of 10 students and perform:

- Mean
- Median
- Maximum value
- Minimum value
- Standard deviation

2. Create a NumPy array from 1 to 20 and:

- Reshape it into a 4×5 matrix
- Find shape of array
- Find dimensions of array

3. Perform array slicing operations:

- First 5 elements
- Last 5 elements
- Alternate elements

4. Create two NumPy arrays and perform:

- Addition
 - Subtraction
 - Multiplication
 - Division
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Part B — Pandas Tasks

5. Load the dataset using Pandas.

6. Display:

- First 5 rows
- Dataset shape
- Column names
- Dataset information

7. Find:

- Total number of passengers
- Total male and female passengers
- Average age
- Maximum fare
- Minimum fare

8. Perform filtering operations:

- Passengers older than 30
- Female passengers
- Passengers who survived
- First-class passengers

9. Check missing values in the dataset.

10. Handle missing values:

- Fill missing Age values with average age
 - Drop rows with missing Embarked values
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Part C — Matplotlib Tasks

11. Create a bar chart showing passenger count by passenger class.

12. Create a pie chart showing male vs female passengers.

13. Create a histogram for Age distribution.

14. Create a scatter plot between Age and Fare.